Лабораторна робота № 4.

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Тема: Взаємне блокування потоків (задача про філософів, що обідають).

Мета: Познайомитись з основними причинами виникнення взаємного блокування паралельних потоків, розібрати способи уникнення блокування потоків на прикладі задачі про філософів, що обідають.

Теоретичні відомості.

Під час розробки паралельних програм, у яких відбувається взаємодія декількох потоків між собою або їх «боротьба» за володіння ресурсами обчислювальної системи, досить часто виникає ситуація, коли один або кілька потоків не можуть продовжити свою роботу через не можливість забезпечення виконання необхідних умов для продовження їх роботи. У такій ситуації говорять про настання тупика або взаємного блокування роботи потоків.

Зазвичай виділяють наступні умови виникнення взаємного блокування потоків: 1) ***умова взаємного виключення*** (*mutual exclusion*). Кожен ресурс у поточний момент або зайнятий рівно одним потоком, або вільний. Тобто, ресурси знаходяться в режимі ексклюзивного користування;

2) ***умова утримання та очікування*** (*hold and wait*). Потоки, що у певний момент утримують отримані раніше ресурси, можуть робити запити на отримання нових ресурсів;
3) ***умова відсутності примусового звільнення ресурсів*** (*no preemption*). Неможливо примусити потік звільнити раніше отримані ресурси. Потік, що володіє ресурсами, повинен сам їх звільняти;
4) ***умова циклічного очікування*** (*circular wait*). Має існувати кільцева послідовність із двох або більше потоків, кожен із яких очікує на звільнення ресурсу, що утримується наступним членом послідовності.

Виконання завдання лабораторної роботи:

Результат роботи програми на мові C# алгоритм Асиметрії та Офіціанта

```
Select way:
1 - AsymmetricWay
2 - WaiterWay
1
Philosopher 3 is thinking.
Philosopher 1 is thinking.
Philosopher 2 is thinking.
Philosopher 0 is thinking.
Philosopher 4 is thinking.
Philosopher 3 picked up first fork 3
Philosopher 3 picked up second fork 4
Philosopher 3 is eatting
Philosopher 2 picked up first fork 2
Philosopher 4 picked up first fork 0
Philosopher 1 picked up first fork 1
Philosopher 3 put down second fork 4
Philosopher 3 put down first fork 3
Philosopher 4 picked up second fork 4
Philosopher 4 is eatting
Philosopher 2 picked up second fork 3
Philosopher 2 is eatting
Philosopher 2 put down second fork 3
Philosopher 4 put down second fork 4
Philosopher 4 put down first fork 0
Philosopher 1 picked up second fork 2
Philosopher 1 is eatting
Philosopher 0 picked up first fork 0
Philosopher 2 put down first fork 2
Philosopher 1 put down second fork 2
Philosopher 1 put down first fork 1
Philosopher 0 picked up second fork 1
Philosopher 0 is eatting
Philosopher 0 put down second fork 1
Philosopher 0 put down first fork 0
```

```
Select way:
1 - AsymmetricWay
2 - WaiterWay
2
Philosopher 1 is thinking.
Philosopher 3 is thinking.
Philosopher 0 is thinking.
Philosopher 4 is thinking.
Philosopher 2 is thinking.
Philosopher 2 couldn't enter the table and will try again later.
Philosopher 4 picked up left fork 4
Philosopher 1 picked up left fork 1
Philosopher 1 picked up right fork 2
Philosopher 1 is eating
Philosopher 3 picked up left fork 3
Philosopher 0 picked up left fork 0
Philosopher 2 is thinking.
Philosopher 1 put down left fork 1
Philosopher 1 put down right fork 2
Philosopher 0 picked up right fork 1
Philosopher 0 is eating
Philosopher 2 picked up left fork 2
Philosopher 4 picked up right fork 0
Philosopher 0 put down left fork 0
Philosopher 0 put down right fork 1
Philosopher 4 is eating
Philosopher 4 put down left fork 4
Philosopher 3 picked up right fork 4
Philosopher 3 is eating
Philosopher 4 put down right fork 0
Philosopher 3 put down left fork 3
Philosopher 3 put down right fork 4
Philosopher 2 picked up right fork 3
Philosopher 2 is eating
Philosopher 2 put down left fork 2
Philosopher 2 put down right fork 3
```

Результат роботи програми на мові Java алгоритм Асиметрії та Офіціанта

```
1
Philosopher 4 is thinking.
Philosopher 2 is thinking.
Philosopher 3 is thinking.
Philosopher 0 is thinking.
Philosopher 1 is thinking.
Philosopher 2 picked up first fork 2
Philosopher 4 picked up first fork 0
Philosopher 2 picked up second fork 3
Philosopher 4 picked up second fork 4
Philosopher 2 is eatting
Philosopher 4 is eatting
Philosopher 1 picked up first fork 1
Philosopher 3 picked up first fork 3
Philosopher 3 picked up second fork 4
Philosopher 3 is eatting
Philosopher 2 put down second fork 3
Philosopher 4 put down second fork 4
Philosopher 1 picked up second fork 2
Philosopher 1 is eatting
Philosopher 0 picked up first fork 0
Philosopher 4 put down first fork 0
Philosopher 2 put down first fork 2
Philosopher 1 put down second fork 2
Philosopher 3 put down second fork 4
Philosopher 0 picked up second fork 1
Philosopher 1 put down first fork 1
Philosopher 0 is eatting
Philosopher 3 put down first fork 3
Philosopher 0 put down second fork 1
Philosopher 0 put down first fork 0
```

```
Philosopher 3 is thinking.
Philosopher 0 is thinking.
Philosopher 4 is thinking.
Philosopher 2 is thinking.
Philosopher 1 is thinking.
Philosopher 0 picked up left fork 0
Philosopher 2 picked up left fork 2
Philosopher 4 picked up left fork 4
Philosopher 0 picked up right fork 1
Philosopher 2 picked up right fork 3
Philosopher 0 is eating
Philosopher 2 is eating
Philosopher 1 couldn't enter the table and will try again later.
Philosopher 4 picked up right fork 0
Philosopher 4 is eating
Philosopher 2 put down left fork 2
Philosopher 0 put down left fork 0
Philosopher 3 picked up left fork 3
Philosopher 2 put down right fork 3
Philosopher 0 put down right fork 1
Philosopher 1 is thinking.
Philosopher 1 picked up left fork 1
Philosopher 1 picked up right fork 2
Philosopher 1 is eating
Philosopher 4 put down left fork 4
Philosopher 4 put down right fork 0
Philosopher 3 picked up right fork 4
Philosopher 3 is eating
Philosopher 1 put down left fork 1
Philosopher 1 put down right fork 2
Philosopher 3 put down left fork 3
Philosopher 3 put down right fork 4
```

Результат роботи програми на мові Kotlin алгоритм Асиметрії та Офіціанта

```
1
Philosopher 0 is thinking.
Philosopher 1 is thinking.
Philosopher 2 is thinking.
Philosopher 3 is thinking.
Philosopher 4 is thinking.
Philosopher 3 picked up first fork 3
Philosopher 3 picked up second fork 4
Philosopher 3 is eating
Philosopher 2 picked up first fork 2
Philosopher 1 picked up first fork 1
Philosopher 0 picked up first fork 0
Philosopher 3 put down left fork 4
Philosopher 3 put down right fork 4
Philosopher 2 picked up second fork 3
Philosopher 2 is eating
Philosopher 2 put down left fork 3
Philosopher 2 put down right fork 3
Philosopher 1 picked up second fork 2
Philosopher 1 is eating
Philosopher 1 put down left fork 2
Philosopher 1 put down right fork 2
Philosopher 0 picked up second fork 1
Philosopher 0 is eating
Philosopher 0 put down left fork 1
Philosopher 0 put down right fork 1
Philosopher 4 picked up first fork 0
Philosopher 4 picked up second fork 4
Philosopher 4 is eating
Philosopher 4 put down left fork 4
Philosopher 4 put down right fork 4
```

```
Philosopher 0 is thinking.
Philosopher 1 is thinking.
Philosopher 2 is thinking.
Philosopher 3 is thinking.
Philosopher 4 is thinking.
Philosopher 4 picked up left fork 4
Philosopher 2 picked up left fork 2
Philosopher 1 picked up left fork 1
Philosopher 3 picked up left fork 3
Philosopher 0 couldn't enter the table and will try again later.
Philosopher 4 picked up right fork 0
Philosopher 4 is eating
Philosopher 0 is thinking.
Philosopher 4 put down left fork 4
Philosopher 3 picked up right fork 4
Philosopher 3 is eating
Philosopher 4 put down right fork 0
Philosopher 0 picked up left fork 0
Philosopher 3 put down left fork 3
Philosopher 3 put down right fork 4
Philosopher 2 picked up right fork 3
Philosopher 2 is eating
Philosopher 2 put down left fork 2
Philosopher 2 put down right fork 3
Philosopher 1 picked up right fork 2
Philosopher 1 is eating
Philosopher 1 put down left fork 1
Philosopher 1 put down right fork 2
Philosopher 0 picked up right fork 1
Philosopher 0 is eating
Philosopher 0 put down left fork 0
Philosopher 0 put down right fork 1
```

Контрольні запитання.

1. Назвіть умови взаємного блокування потоків.

2. Яким чином можна подолати умову взаємного виключення?

3. Яким чином можна подолати умову утримання та очікування?

4. Яким чином можна подолати умову відсутності примусового звільнення ресурсів?

5. Яким чином можна подолати умову циклічного очікування?

Посилання на Git-репозиторій: https://github.com/OleksandrTretiak1/LabPP4